Rank-27 Vivify Parity Repair

Tactix engineering handoff

21 August 2026

Objective and outcome

The requested audit covered Vivify admission, scheduling, candidate and literal order, interaction with reason jumping, terminal proof rules, and interleaving with the surrounding inprocessing pipeline. The concern was correct: Tactix carried an invented size-100 candidate cutoff absent from both reference solvers. Removing it exposed three additional coupled differences from Kissat: independent rather than cumulative tier budgets, premature expansion of a jumped binary reason during semantic deduction, and a replacement-only productivity numerator that delayed away the next irredundant round.

The production repair follows the reference mechanism without input identity or tuning. It removes the cutoff, charges schedule and watch rebuilding to a single cumulative Vivify ceiling, reads the virtual jumped reason for semantic subsumer detection while retaining real-antecedent proof expansion, and counts every successful terminal in productivity. The final common-state trace restores Kissat's second irredundant round, exact long-clause counts, and exact aggregate terminal results. Every binary-producing probe terminal also has the same effect.

Release passes 72/72 tests. The mandatory 28B+5C matched guard is 66/66 valid; all models and proofs check and no certificate remains. This is a semantic and correctness closure, not yet a performance-pin acceptance: the candidate takes 1.241 times the accepted pin's aggregate time. The development plan therefore treats the timing regression as evidence of a still-missing coupling and forbids restoring the hack.

Frozen identities and experiment discipline

The immutable expected-UNSAT rank-27 input is `tierD24-26/5911653e4edadec391654c8f06353f9a-oski15a01b06s_opt.cnf.xz`, compressed SHA-256 `5435b1b2...bb4e9`. The accepted binary is `b6c991e0...fb6f`, source `7b527fbc...b4b`. The repaired candidate is `df9c9805...9c3b`, built from base `1ba9f021...7f6` plus source diff `37d8fffa...d912`.

The branch and `origin/main` were synchronized before work. Surviving worktrees and SLURM jobs were inspected and left untouched. The frozen rank-21–40 manifest, prior closures, and retained reports were read before new experiments. No frozen reference performance solver was rerun; Kissat was used only through retained traces and pinned, diagnostic-only source. Every real solver run used SLURM. The required debug contract was two CPUs, 16 GiB, and a 1,200-second solver timeout; diagnostic traces used 16 GiB and 300 seconds.

The four source-backed defects

At conflict 3069, the retained traces have a common semantic formula and equal long/binary populations before Vivify.

Candidate scope. Tactix's first skipped changed candidate had 435 literals. The only cause was `VIVIFY_CLSLIM=100`. Kissat `vivify.c` has no candidate-size test; its propagation-tick ceiling bounds work. After deleting the cutoff, the first 32 candidate identities, priorities, order keys, and probe poses match. A 101-literal focused regression prevents the invented policy from returning.

Cumulative scheduling. Kissat computes one absolute probing-tick allowance, then advances its ceiling by the 3/3/1 redundant shares and the irredundant share. Unused early work is available later, while an early overrun is precharged. Tactix gave every band a fresh independent budget and started its clock after schedule/watch work. Two policy fields now carry the already-spent offset and distinguish bounded zero from an ordinary unbounded call. A synthetic tier-3 fixture requires unused tier-1/tier-2 budget to carry.

Jumped semantic reason. The first terminal split was candidate hash `1c3734085b121754`, literals `(-480909,480954,480955)`. Kissat subsumed it; Tactix rewrote it to a binary. The assignment chain was `480909:binary(96248) | 96248:binary(480954) | 480954:decision`. Reason jumping stores the virtual resolvent `(-480909 or 480954)`. Tactix expanded through 96248 before semantic subsumer detection, hiding that resolvent. Deduction now reads the virtual reason; native proof walks still expand both real antecedents. A focused test checks both the subsumption and the proof-safe separation.

Productivity and interleaving. After the semantic repair, Tactix still ran seven Vivify rounds while Kissat ran eight. Its irredundant yield counted replacement clauses only, omitting 7,982 subsumptions. That raised the delay and skipped the next irredundant band. The numerator now includes subsumption, shrink, implied/asymmetric deletion, and instantiation, matching Kissat's per-candidate `vivifier.vivified`. A subsumption-only scheduling regression requires the delay to fall rather than rise.

Final semantic accounting

Trace job `1637436` contains both Vivify calls. At the end of the second irredundant round, both solvers have exactly 574,378 irredundant and 479 redundant long clauses. Aggregate outcomes are also exact:

	subsumed	shrunken	implied	asymmetric	instantiated
Tactix	7,982	182	323	0	17
Kissat	7,982	182	323	0	17

The snapshot has 34 more Tactix binaries after both calls. Bounded diagnostic jobs `1637438`, `1637440`, and `1637441` separate that count: 21 belong to the already-separate TransitiveReduce call between Vivify rounds, and 13 are schedule-time cleanup of Tactix's inherited retained-root representation. All 69 binary-producing probe terminals match in effect. Five candidates take a different terminal label—strict shrink in Tactix, binary instantiation in Kissat—but emit identical

clauses. For candidate `b34e59b48820cebb`, both emit `(-435735 or -435727)`; the other four pairs also match exactly. This distinguishes a counter/path label from a formula-effect divergence.

Temporary diagnostic output was compiled only into immutable trace pins and removed from production source before final validation.

Correctness and performance evidence

The clean Release build passes all 72 tests, including native proof checks and the five new Vivify/scheduling regressions. No test, fuzz evidence, or campaign exposed a crash or soundness defect, so the plan's mandatory Claude repair-delegation trigger did not fire.

Matched job `1637369`, campaign `benchmarks/runs/20260821-1300-BC-vivify-productivity/`, ran all 28 Tier-B fixtures and the inherited fixed five mechanism-relevant Tier-C fixtures on both pins, without a node constraint or array throttle.

Arm	valid	SAT / UNSAT	aggregate	peak RSS
Accepted pin	33/33	17 / 16	2233.697 s	3.233 GiB
Candidate	33/33	17 / 16	2772.060 s	3.098 GiB

All 34 SAT models and 32 UNSAT proofs verify. The aggregate reports no invalid, malformed, missing, cleanup-bad, or residual-certificate task. The ratio is 1.241019. The largest named remaining witnesses are Tier-C `mp1-Nb5T15` (712.115 versus 427.820 seconds) and Tier-B `sted5` (199.798 versus 36.035 seconds); `mod4block` is 109.431 versus 93.951 seconds. These numbers reopen coupling localization. They do not justify a size cutoff, formula classifier, or per-instance policy.

Rank-27 witness

Only after the matched guard passed was immutable rank-27 job `1637437` submitted under the 16 GiB/1,200-second contract. It returned UNKNOWN at 1,200.004 seconds, so rank 27 is not added to the closed set. The runner recorded a 252,706,816-byte partial proof (SHA-256 `c2566396...d350`), deleted it and all derived scratch, and published zero certificate residue. Scheduler peak RSS was 453,336 KiB. Evidence is `benchmarks/runs/20260821-1430-tierd-rank27-vivify-parity/`.

Limitations and next action

The repaired code closes the identified Vivify semantics at common-input candidate, schedule, deduction, and emitted-binary layers. It does not claim that every later rank-27 search transaction matches Kissat, nor that the candidate is a new accepted performance pin. If rank 27 remains open, the next UNSAT-first action is to continue from the first portable post-Vivify divergence, without repeating candidate-scope, cumulative-budget, jumped-reason, or productivity investigations. Separately, the predeclared `mp1-Nb5T15`/`sted5` regression must be localized as a missing coupling before performance acceptance. Any further source change again owes the complete 28B+5C guard before another Tier-D witness.

The durable technical report is `benchmarks/reports/2026-08-21-tierd-rank27-vivify-parity.md`. Current handoff is in `PROGRESS.md`; durable mechanism evidence is in `docs/implementation-notes.md` and `docs/scheduling-parity-ledger.md`.